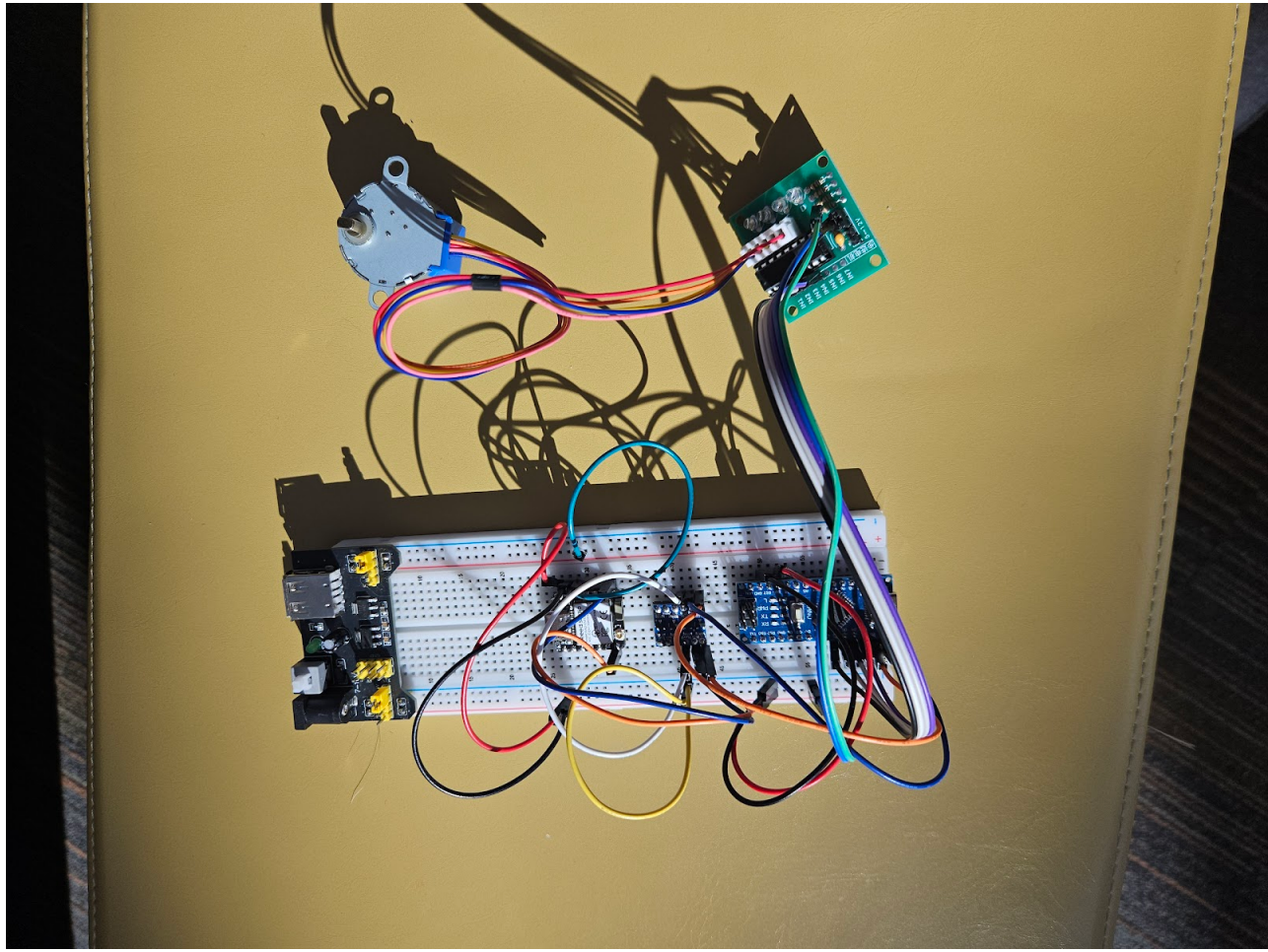
S1P - Jacob Canada

Current prototype

Components:

- ESP32C6
- Level shifter
- Arduino Nano
- 28BYJ-48 stepper motor with ULN2003 driver board
- MB102 power supply module



The setup

- Laptop and ESP32C6 connected to phone hotspot
- Send an HTTP POST request with the "steps" parameter set to the desired number of motor steps
- Arduino receives that value over serial and turns the motor
- https://drive.google.com/file/d/1tAgiaYA7ABROQSSkut-7g0KS_YXNRxTX/preview

Learning with AI Topics

- I have learned how to use stepper motors, which covers a hardware and software topic.
- I still need to learn user IO in the form of buttons and a display; I need to figure out what I need and how to use it in software.

Plans for Sprint 2

- Fix unresolved issue - Arduino still needs a USB connection despite the external power supply
- Mount motor to my telescope's RA axis and calculate the rate the motor needs to spin to track an object
    - This would be all that is necessary for simple object tracking.
- Integrate a display (LCD/OLED?) and buttons to engage/disengage tracking and inform the user what is going on
- If time allows, I will add a second motor and implement the full go-to functionality.
    - This is why I added the ESP32C6 for my S1 prototype; I wanted a way to send numerical data to the Arduino.